

2.1 TR & TE — THE FOUNDATION OF CONTRAST

TWO SIMPLE KNOBS THAT SHAPE ALMOST EVERY IMAGE

Repetition Time (TR) and Echo Time (TE) are the most powerful parameters for controlling image contrast. They determine how much longitudinal (T1) and transverse (T2) relaxation occurs before the signal is measured.

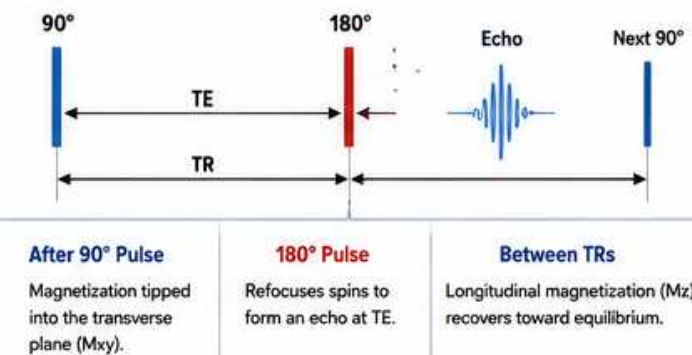


KEY CONCEPT

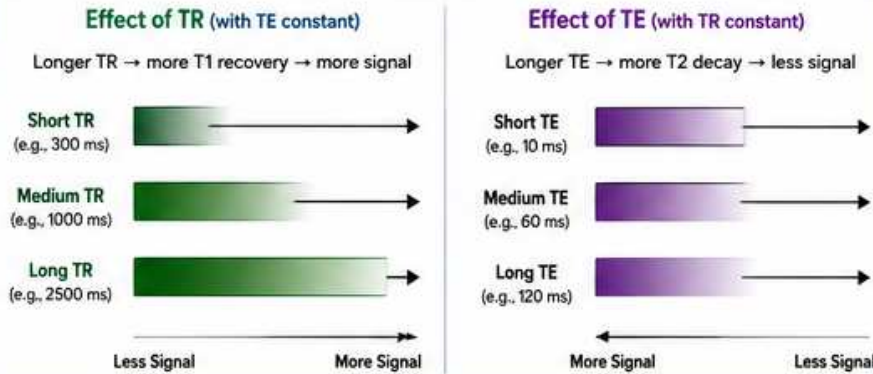
TR controls recovery (longitudinal magnetization).
TE controls decay (transverse magnetization).
Different combinations create different tissue contrast.

↑ Increase
↓ Decrease
— No / Minimal Effect

A. WHAT HAPPENS IN ONE PULSE SEQUENCE (SPIN ECHO EXAMPLE)



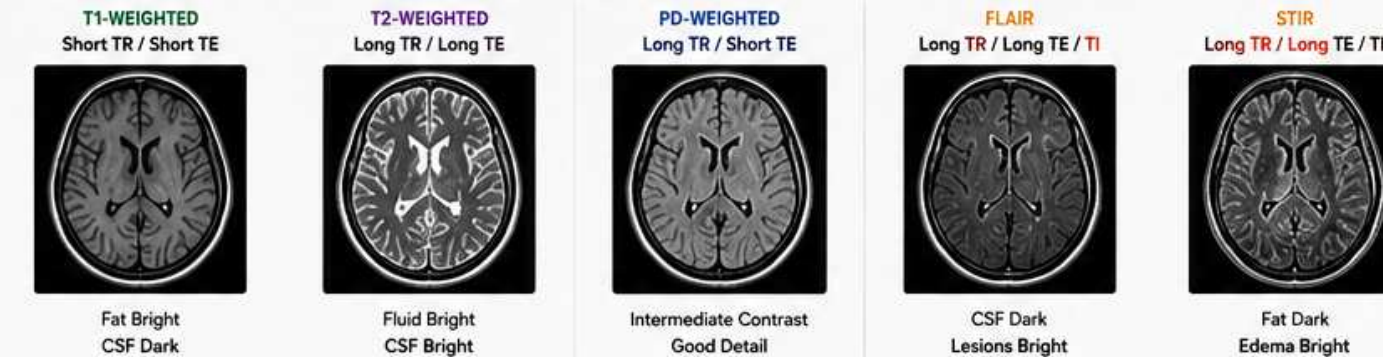
B. HOW TR & TE AFFECT SIGNAL



C. CONTRAST RESULTING FROM TR & TE CHOICES

Image Type	TR	TE	T1 Contrast	T2 Contrast	Typical Use
T1-Weighted	Short	Short	↑	↓	Anatomy, post-contrast, fat vs water
T2-Weighted	Long	Long	↓	↑	Fluid, edema, pathology
PD-Weighted	Long	Short	—	—	Disc, tendon, brain, joint evaluation
FLAIR	Long	Long	↓	↑	Suppress CSF, lesions near fluid
STIR	Long	Long	↓	↑	Fat suppression, edema, inflammation
Inversion Recovery	Varies	Short	↑	↓	Tissue nulling (e.g., T1 for FLAIR, STIR)

D. VISUAL EXAMPLES — SAME ANATOMY, DIFFERENT TR & TE



NOTE

Images shown are illustrative. Actual appearance varies with anatomy, field strength, coils, and other parameters.

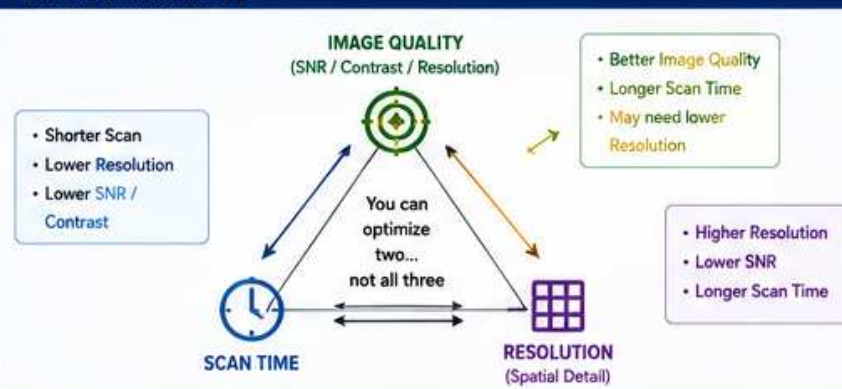
E. TROUBLESHOOTING: WHEN IMAGES DON'T LOOK RIGHT

Observation	Likely Cause	Check / Action
Image too dark	TR too short or TE too long	Increase TR, decrease TE, increase NEX, check coil and prescan
Grainy / noisy image	Low SNR (short TR, low NEX, thin slice)	Increase NEX, increase TR, thicker slice, optimize BW
Poor T1 contrast	TR too long	Shorten TR
Poor T2 contrast	TE too short	Increase TE
Scan time too long	Long TR, high NEX, high matrix, long TE	Reduce NEX, use parallel imaging, lower matrix, shorten TR/TE if possible

F. AT A GLANCE: PARAMETER EFFECTS

Parameter	↑ Increase ↑	↓ Decrease ↓	Main Trade-offs
TR	• ↑ SNR • ↑ T1 weighting (less) • ↑ Scan time	• ↓ SNR • ↑ T1 weighting (more) • ↓ Scan time	SNR vs T1 contrast vs Scan Time
TE	• ↑ T2 weighting (more) • ↓ Signal (T2 decay) • ↑ Scan time	• ↓ T2 weighting (less) • ↑ Signal • ↓ Scan time	T2 contrast vs Signal vs Scan Time
NEX/NSA	• ↑ SNR • ↑ Scan time	• ↓ SNR • ↓ Scan time	SNR vs Scan Time
Bandwidth	• ↓ Distortion • ↑ Noise (↓ SNR)	• ↑ SNR • ↑ Distortion (EPI)	SNR vs Distortion

G. THE MRI TRIANGLE



H. ENGINEER CHECKLIST

- ✓ Always ask: What was the TR and TE?
- ✓ Compare with a known good protocol.
- ✓ Is the protocol intentionally optimized for speed or detail?
- ✓ What is the clinical goal (T1, T2, PD, FLAIR, etc.)?
- ✓ Do artifacts follow parameter choices or hardware issues?
- ✓ Always verify prescan data and coil performance.



KEY TAKEAWAY

TR and TE are the primary controls of image contrast. Understanding their impact helps the engineer determine whether an issue is protocol-related or system-related.